

Interpreting Graph Neural Networks with MIPExplainer: Sparse Explanations for Alzheimer’s and Autism

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Abstract—Classifying neurological conditions using functional magnetic resonance imaging (fMRI) remains a challenging task due to the complexity of brain connectivity patterns. We address this by modeling brain networks as graphs, where nodes represent brain regions and edges represent functional connectivity derived from blood-oxygen-level-dependent (BOLD) signals. Graph neural networks (GNNs) are trained on these graphs to distinguish patients with Alzheimer’s disease (AD) and Autism Spectrum Disorder (ASD) from healthy controls. To enhance interpretability, we introduce a Mixed Integer Programming (MIP)-based framework that identifies discriminative subgraphs and features driving model decisions. We preprocess data using the fMRIPrep pipeline and construct connectivity graphs with the ACTION toolbox, utilizing datasets from the Alzheimer’s Disease Neuroimaging Initiative (ADNI) and the Autism Brain Imaging Data Exchange (ABIDE).

I. INTRODUCTION

Understanding the intricate patterns of brain activity is critical for diagnosing and characterizing neurological disorders such as Alzheimer’s Disease (AD) and Autism Spectrum Disorder (ASD). Functional magnetic resonance imaging (fMRI), which captures blood-oxygen-level-dependent (BOLD) signals, has emerged as a powerful modality for exploring the brain’s functional connectivity. These BOLD signals reveal complex interactions among brain regions that can naturally be modeled as graphs, where nodes represent regions of interest (ROIs) and edges encode functional associations between those regions.

While recent advances in graph neural networks (GNNs) have enabled accurate classification of neurological conditions from such graph-structured data, their black-box nature limits clinical interpretability. Understanding not only the predictions of a model, but also why, is essential for aligning computational findings with known neurobiological mechanisms. In this study, we focus on addressing this interpretability gap using MIPExplainer [1], an optimization-based method that generates faithful, instance-level explanations for GNN predictions through mixed integer programming.

We begin by preprocessing resting-state fMRI data from the Alzheimer’s Disease Neuroimaging Initiative¹ [2] (ADNI) and the Autism Brain Imaging Data Exchange (ABIDE) [3] using the fMRIPrep [4] pipeline. Functional brain graphs are constructed using the ACTION toolbox [5], with nodes representing ROIs and edges defined by measures such as Pearson’s correlation or sparse representation. GNNs are then trained to classify each subject as a patient or control.

To interpret these predictions, we apply MIPExplainer [1] to identify the minimal set of brain regions and connections whose perturbation changes the model’s output. This approach yields sparse, biologically plausible subgraphs that highlight the most discriminative features for each classification. By systematically analyzing the explanations across subjects and cohorts, we gain insight into disorder-specific connectivity patterns in AD and ASD, bridging predictive modeling with mechanistic understanding.

II. RELATED WORK

A. Graph Neural Networks (GNNs)

GNNs have become a powerful framework for modeling complex relational data, making them well-suited to analyze functional brain networks derived from fMRI scans. In these networks, brain regions are represented as nodes and functional connections (based on statistical associations such as Pearson’s correlation, sparse representation, partial correlation, etc.) are encoded as edges. GNNs enable the learning of representations that capture complex patterns of connectivity while accounting for the underlying graph structure.

Several prior works have explored the application of GNNs for brain disorder classification. For example, the FC-HAT

¹Data used in preparation of this article were obtained from the Alzheimer’s Disease Neuroimaging Initiative (ADNI) database (adni.loni.usc.edu). As such, the investigators within the ADNI contributed to the design and implementation of ADNI and/or provided data but did not participate in analysis or writing of this report. A complete listing of ADNI investigators can be found here: http://adni.loni.usc.edu/wp-content/uploads/how_to_apply/ADNI_Acknowledgement_List.pdf

model [6] introduced a hypergraph attention mechanism to account for high-order interactions among brain regions and achieved strong performance on datasets including ABIDE and ADNI. Similarly, Xiao et al. proposed a multi-hypergraph learning framework [7] that constructs functional connectivity networks using sparse representation and adaptive hyperedge weighting. This approach demonstrated improved discriminability in learning ability classification tasks using multi-paradigm fMRI data.

Other studies have focused on improving the structure of the functional networks themselves. For instance, Li et al. remodeled Pearson’s correlation as an optimization problem with sparsity constraints to create more biologically plausible, sparse networks. [8] This method yielded improved performance in identifying individuals with ASD.

While these methods have advanced the classification of neurological disorders using GNNs, they often lack interpretability. Most GNN-based approaches function as black boxes, making it difficult to understand which features or subgraphs drive the model’s predictions. Our work addresses this gap by applying MIExplainer [1], a mixed integer programming-based framework for generating post hoc, model-agnostic explanations of GNN decisions. By combining high-performing GNNs with MIP-based interpretability, we aim to uncover meaningful neural biomarkers associated with Alzheimer’s Disease and Autism Spectrum Disorder, while preserving transparency in the decision process.

B. GNN Explainability

As GNNs have become ubiquitous in a variety of applications, understanding their predictions has become increasingly important. Significant work in the domain of explainable GNNs has been dedicated to creating post-hoc explanations. GNNExplainer [9] pioneered instance-level explanations for GNNs by identifying critical subgraphs using continuous mask optimization. Later approaches, such as PGExplainer [10], further improved efficiency and scalability by parameterizing edge selection with neural predictors. SubgraphX [11] explored explanations through Monte Carlo Tree Search (MCTS), providing explanations that explicitly optimize predictive fidelity and conciseness. MIExplainer [1] [12], to which this paper is an extension of, suggests utilizing mixed integer programming to explain GNNs at an instance level. We expand upon this work by providing explanations to brain networks.

C. Mixed-Integer Programming for Explainability

Recent research has explored the use of optimization frameworks, specifically Mixed-Integer Programming (MIP), to rigorously enforce constraints on explanatory structures, providing interpretable and precise explanations. [1] Gaines et al. introduced a MIP-based framework for explaining GNN predictions, showing that optimization techniques can produce more accurate and interpretable explanations compared to heuristic-based methods. [1] This work demonstrated the

effectiveness of constrained optimization to identify minimal and meaningful substructures within graphs.

D. Main Contributions

In this paper, we seek to identify and interpret functional connectivity patterns in the brain that are predictive of AD and ASD. To do this, we preprocess resting-state fMRI data to construct subject-specific brain graphs, train Graph Neural Network (GNN) classifiers to distinguish between patients and controls, and apply MIExplainer to extract sparse, model-faithful explanations for individual predictions. This section describes the steps involved in data preparation, graph construction, model training, and explanation generation in detail.

III. METHOD

A. Preprocessing Brain Data

Functional MRI data were preprocessed using fMRIPrep 25.1.3 [13], [14], a robust and standardized pipeline built on Nipype [15] and designed to maximize reproducibility. The preprocessing steps included both anatomical and functional workflows, as outlined below.

Anatomical preprocessing began with skull-stripping of the T1-weighted (T1w) image, followed by spatial normalization to the MNI152NLin2009cAsym template using nonlinear registration (ANTs) [16]. Brain tissue was segmented into gray matter (GM), white matter (WM), and cerebrospinal fluid (CSF) using the FAST algorithm from FSL [17]. Surface reconstruction was performed with FreeSurfer’s recon-all, and the anatomical-to-standard-space transformation was applied to functional data.

Functional preprocessing included slice timing correction, head motion correction (MCFLIRT) [18], and co-registration of the BOLD reference image to the T1w image using FreeSurfer’s bbregister with boundary-based registration [19]. Functional images were resampled to MNI152 space and to the original native space for further analysis [20]–[22]. To account for physiological and motion-related noise, a set of nuisance regressors was extracted, including:

- Six head motion parameters and their derivatives
- Framewise displacement (FD) and DVARS
- Global signal, WM, and CSF mean signals
- Anatomical CompCor (aCompCor) components estimated from WM and CSF masks
- Cosine drift terms for high-pass filtering

The correlation structure among these regressors (Fig. 3, Page 4) was inspected to identify redundancy and guide selection for nuisance regression. A carpet plot (Fig. 2, Page 3) of the BOLD time series was also generated to visualize temporal structure and motion-related artifacts.

All transformations (motion correction, susceptibility distortion correction, co-registration, and normalization) were combined into a single interpolation step using Lanczos resampling to minimize data smoothing.

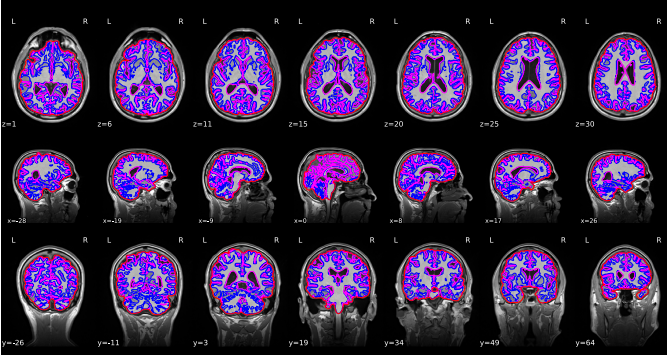


Fig. 1: This figure shows the final, preprocessed T1-weighted image, with contours delineating the detected brain mask and brain tissue segmentations for one subject.

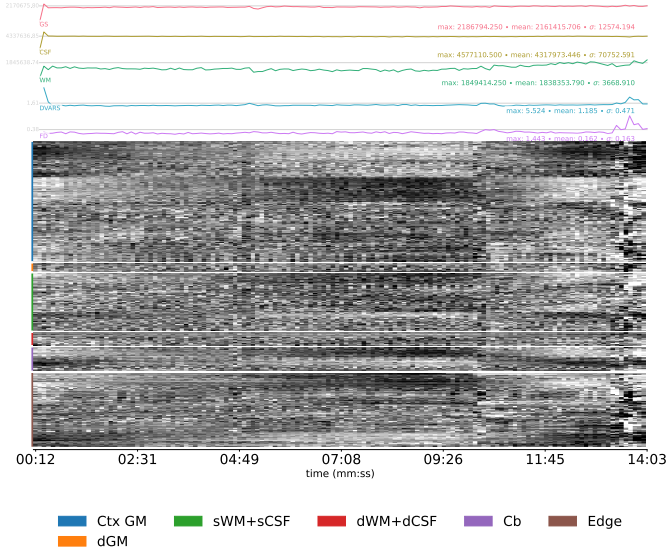


Fig. 2: A carpet plot shows the time series for all voxels within the brain mask for one subject. See the figure legend for specific color mappings. "Ctx" = cortex, "Cb" = cerebellum, "WM" = white matter, "CSF" = cerebrospinal fluid. "d" and "s" prefixes indicate "deep" and "shallow" relative to the cortex. "Edge" indicates regions just outside the brain.

B. Brain Network Construction

Following preprocessing, we construct subject-specific functional brain networks using the *ACTION* toolbox [5]. For each subject, the preprocessed BOLD time series are first parcellated into n regions of interest (ROIs) according to the Harvard–Oxford cortical and subcortical atlas. *ACTION* then provides a modular pipeline that (i) standardizes each ROI’s time series, (ii) computes pairwise association measures, and (iii) assembles these measures into an $n \times n$ connectivity matrix.

We adopt two association metrics supplied by the *ACTION* toolbox: Pearson’s Correlation (PC) and Sparse Representation (SR), explained below. The toolbox thresholds or regularizes these matrices to obtain binary or weighted adjacency matrices

while preserving graph sparsity and connectedness. All resulting graphs are stored in standard edge-list format and was used to train our GNN models for classification.

1) *Pearson’s Correlation (PC)*: PC measures the linear relationship between two BOLD time series, reflecting how synchronized the activity of two regions is over time. Given two time series X and Y from different brain regions, Pearson’s correlation coefficient is defined as:

$$PC_{XY} = \frac{\text{cov}(X, Y)}{\sigma_X \sigma_Y} \quad (1)$$

where $\text{cov}(X, Y)$ is the covariance between X and Y , and σ_X, σ_Y are their standard deviations. The resulting value ranges from -1 to $+1$, with higher absolute values indicating stronger linear coupling in corresponding directions. When computed across all region pairs, these correlations form a functional connectivity matrix that can be interpreted as a fully connected brain graph.

2) *Sparse Representation (SR)*: Sparse Representation (SR) estimates brain functional networks by modeling each region’s activity as a sparse linear combination of other regions.

The SR model solves the following optimization problem looking to balance the accuracy of $W \cdot X$ and the sparsity of W :

$$\min_W \{ \|X - XW\|_2^2 + \lambda \|W\|_1 \} \quad (2)$$

where $X \in \mathbb{R}^{N \times T}$ is the fMRI data matrix, $W \in \mathbb{R}^{N \times N}$ is the learned sparse connectivity or coefficient matrix, and $\lambda \in [0, \infty)$ controls the sparsity. High values of λ yield more sparse graphs while λ values approaching 0 allow for more connections to be made.

C. Explaining Predictions with MIPEXPLAINER

To interpret the predictions made by our trained Graph Neural Network (GNN) models, we utilize MIPEXPLAINER [1], an optimization-based method that provides model-agnostic, instance-level explanations using Mixed Integer Programming (MIP). This technique is designed to identify the minimal subset of nodes and edges in a graph whose perturbation changes the model’s output, thus offering insight into the most influential brain regions and connections for each prediction.

$$f_\theta(G) = y, \quad f_\theta(G') = y, \quad \forall y \in \mathcal{Y} \quad (3)$$

while minimizing the number of retained edges. For $z \in \mathbb{R}^{|E| \times |E|}$, we minimize:

$$\min_z \|z_{a,b}\|_0, \quad \text{where } z = 1 \iff \text{edge}(a, b) \in E' \quad (4)$$

G' is subject to constraints that ensure interpretability and validity of the perturbed graph structure.

In our experiments, we apply MIPEXPLAINER [1], [12] to functional connectivity graphs derived from fMRI data, which were preprocessed with fMRIPrep [13], [14] and constructed using the *ACTION* toolbox [5]. The datasets include subjects from the Alzheimer’s Disease Neuroimaging Initiative (ADNI) [2] and the Autism Brain Imaging Data Exchange (ABIDE) [3]. Each subject’s brain is represented as a graph where

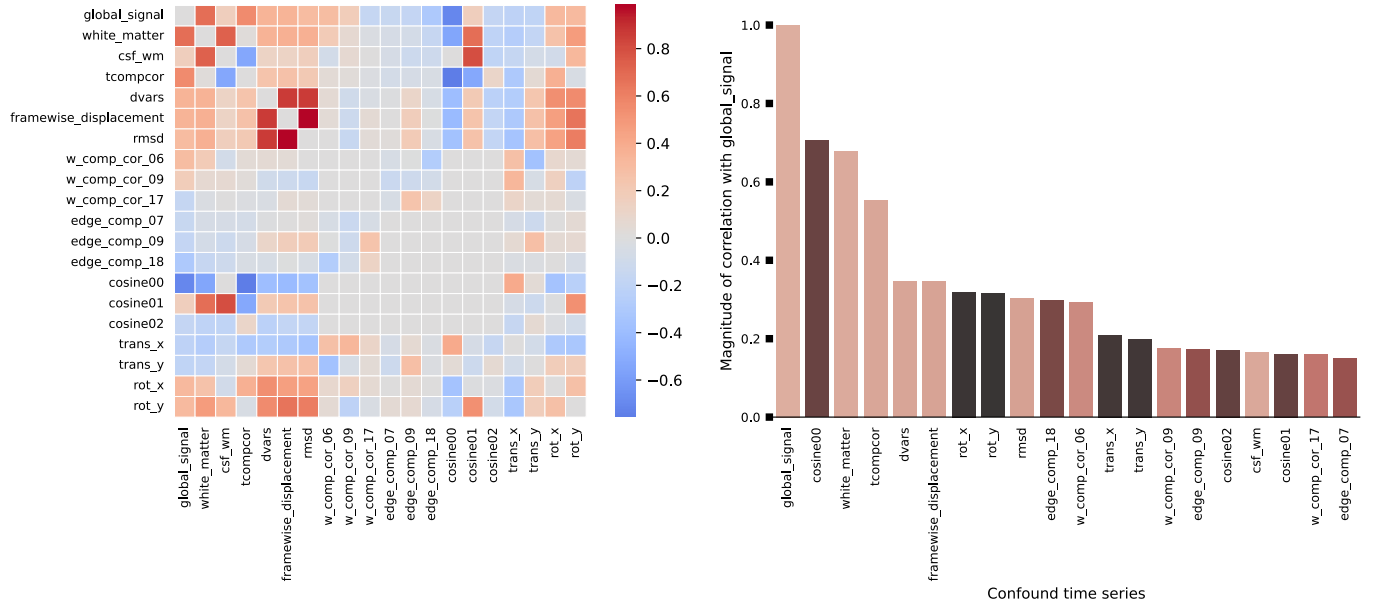


Fig. 3: Left: Heatmap summarizing the correlation structure among confound variables for one subject. (Cosine bases and PCA-derived CompCor components are inherently orthogonal.) Right: magnitude of the correlation between each confound time series and the mean global signal. Strong correlations might be indicative of partial volume effects and can inform decisions about feature orthogonalization prior to confound regression.

nodes correspond to anatomical regions of interest (ROIs), and edges encode functional connectivity based on BOLD signal correlations. We trained GNNs to classify each subject as either a patient or control, and then used MIPEXplainer to generate explanations for a subset of these predictions.

The explanations returned by MIPEXplainer consist of a sparse set of nodes and edges deemed most critical for the model’s classification. For example, in cases where the GNN correctly classified a subject as having Alzheimer’s Disease, MIPEXplainer frequently identified disrupted connectivity between regions in the default mode network (DMN), consistent with existing literature on AD biomarkers. An example explanation graph can be found in Fig. 4, Page 5.

This method produces sparse, targeted explanations that align with the internal logic of the GNN and support biological interpretability. Unlike gradient-based saliency maps [23] or attention mechanisms [24], which often suffer from sensitivity and inconsistency, MIPEXplainer guarantees faithfulness to the model by directly modeling its activation patterns within an exact optimization framework. This makes it particularly suitable for scientific domains like neuroimaging, where interpretability is crucial.

To facilitate the formulation of the MILP, we encode each layer of the GNN using a combination of continuous variables for the activations and binary variables for ReLU thresholds. For models with multiple layers and non-linear activations, the number of variables and constraints grows quickly, so we limit the number of layers and hidden units where possible and freeze the network parameters during explanation generation. The encoding process is model-agnostic and can be adapted

to GCN [25], GAT [26], GraphSAGE [27], and other architectures. In our work, we apply MIPEXplainer to models trained with GraphSAGE layers.

We also introduce mild constraints during optimization to improve the plausibility of generated explanations. For example, we prevent disconnection of key network components, limit the number of removed or added edges, and preserve degree distributions within realistic ranges. These constraints help ensure that the resulting explanation corresponds to a biologically feasible perturbation rather than a pathological graph configuration.

Once explanations are generated, we visualize the identified subgraphs and analyze the frequency with which specific regions and connections appear across samples. This allows us to identify not only which features are important for individual predictions but also which patterns are consistently influential across subjects. We further compare explanations across AD and ASD cohorts to examine whether the discriminative features vary across disorders or reflect shared mechanisms.

To validate the informativeness of MIPEXplainer outputs, we compute explanation sparsity, fidelity (i.e., how confidently the model’s prediction changes when the perturbation is applied), and overlap with known biomarkers reported in the neuroimaging literature. In several high-confidence cases, the nodes and edges highlighted by MIPEXplainer correspond to regions known to be impacted by neurodegenerative or neurodevelopmental conditions, supporting the clinical relevance of our findings.

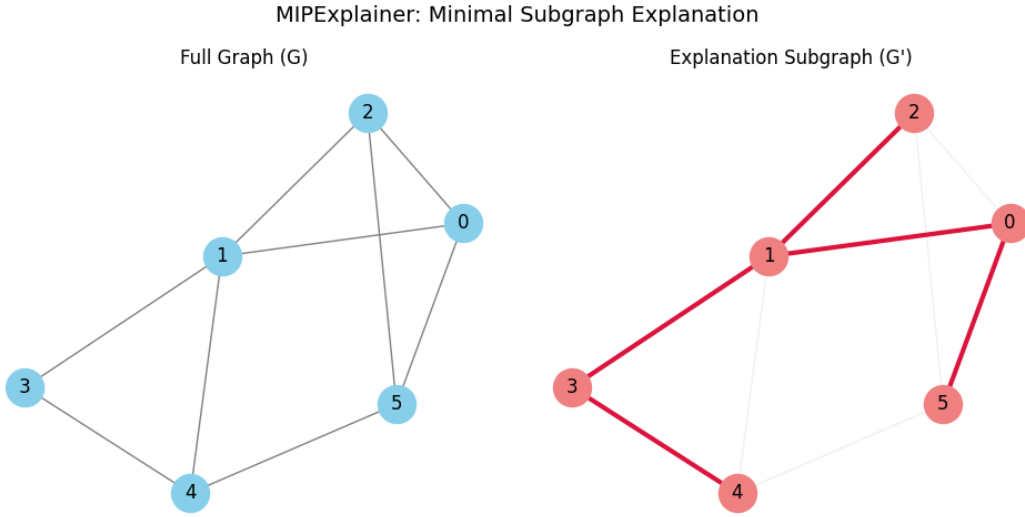


Fig. 4: This figure illustrates how MIPEXplainer identifies a minimal subgraph that preserves the original GNN prediction. On the left is the full input graph G , and on the right is the explanation subgraph G' , containing only the critical edges (in red) necessary to maintain the same predicted class. This sparse explanation highlights the most influential structure behind the model’s decision.

IV. EXPERIMENTS

A. Datasets

1) *ADNI Dataset*: Data used in the preparation of this article were obtained from the Alzheimer’s Disease Neuroimaging Initiative (ADNI) database [2] (<https://adni.loni.usc.edu>). The ADNI was launched in 2003 as a public-private partnership, led by Principal Investigator Michael W. Weiner, MD. The primary goal of ADNI has been to test whether serial magnetic resonance imaging (MRI), positron emission tomography (PET), other biological markers, and clinical and neuropsychological assessment can be combined to measure the progression of mild cognitive impairment (MCI) and early AD.

2) *ABIDE Dataset*: The Autism Brain Imaging Data Exchange (ABIDE) [3] is a publicly available multi-site dataset that aggregates resting-state fMRI and anatomical MRI scans from individuals with Autism Spectrum Disorder (ASD) and typically developing controls. The dataset includes subjects ranging from children to adults, collected across over 20 international sites, providing a diverse and heterogeneous sample.

B. Results

1) *fMRIPrep*: Results included in this manuscript come from preprocessing performed using *fMRIPrep* 25.1.3 [13], [14], RRID:SCR_016216, which is based on *Nipype* 1.10.0 [28], [29], RRID:SCR_002502.

V. CONCLUSION

In this study, we present an interpretable framework for classifying neurological conditions using Graph Neural Networks applied to functional brain networks constructed from fMRI

data. Leveraging the ACTION toolbox for standardized graph construction and MIPEXplainer for post-hoc explanation, our approach provides sparse, faithful insights into the underlying functional connectivity patterns. MIPEXplainer enables us to perturb the graph structure in an optimization-driven way, identifying the most influential brain regions and edges for each individual prediction.

This work demonstrates the potential of combining machine learning with discrete optimization to enhance transparency in clinical neuroimaging. By systematically applying MIPEXplainer across a diverse set of subjects, we uncover interpretable patterns of altered connectivity that are consistent with known biomarkers, offering a bridge between data-driven models and neuroscientific understanding. Our approach supports not just high performance, but meaningful biological interpretation — a critical step toward the adoption of GNN-based tools in real-world diagnostic settings.

Future work will explore generalization to other brain disorders such as Parkinson’s disease, expand to hypergraph neural networks and explanations, and incorporate multi-modal neuroimaging data to improve both predictive power and the biological fidelity of explanations.

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